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SYNTHESIS OF NANOSTRUCTURED MATERIALS BY MECHANICAL MILLING: PROBLEMS AND OPPORTUNITIES

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Abstract *— Mechanical attrition as a method to produce nanocrystalline (nc) materials is reviewed. Its advantages include the fact that all classes of materials — including brittle compounds — are amenable to the method; it can be easily scaled up to tonnage quantities. The phenomenology and suggested mechanisms for formation of nc microstructures are discussed for ball milling of single component powders, mechanical alloying of multi-component powders, and mechanical crystallization of amorphous alloys. The phenomenology is well documented but microscopic mechanisms await better understanding of the nature of deformation processes in nc materials. The problems of contamination and powder consolidation are briefly considered. ©1997 Acta Metallurgica Inc.*

INTRODUCTION

A wide variety of techniques are being used to synthesize nanostructured materials including inert gas condensation (1) rapid solidification (2), electrodeposition (3), sputtering (4), crystallization of amorphous phases (5), and chemical processing (6). Mechanical attrition — ball milling — which induces heavy cyclic deformation in powders, is a technique which has also been used widely for preparation of nanostructured materials (7). Unlike many of the above methods, mechanical attrition produces its nanostructures not by cluster assembly but by the structural decomposition of coarser-grained structures as the result of severe plastic deformation. This has become a popular method to make nanocrystalline materials because of its simplicity, the relatively inexpensive equipment (on the laboratory scale) needed, and the applicability to essentially all classes of materials. The major advantage often quoted is the possibility for easily scaling up to tonnage quantities of material for various applications. Similarly, the serious problems that are usually cited are 1) contamination from milling media and/or atmosphere, and 2) the need (for many applications) to consolidate the powder product without coarsening the nanocrystalline microstructure. In fact, the contamination problem is often given as a reason to dismiss the method, at least for some materials.

This paper will review the mechanisms presently believed responsible for formation of nanocrystalline structures by mechanical attrition of single phase powders, mechanical alloying of dissimilar powders, and mechanical crystallization of amorphous materials. The two important problems of contamination and powder consolidation will be briefly considered.

THE EVOLUTION OF NANOCRYSTALLINE STRUCTURES IN SINGLE PHASE MATERIALS BY MECHANICAL ATTRITION

The present understanding of the development of a nanocrystalline microstructure by ball milling has been reviewed previously (7,8,9). This information comes mainly from a TEM and high resolution TEM (HREM) study of ball-milled Ru and AlRu microstructures as a function of milling time (10,11). Using x-ray line broadening to measure "grain" size it is commonly observed that the crystallite size decreases with milling time. In the TEM and HREM study (11) of AlRu it was observed that at the early stages of milling the deformation was localized within shear bands that are approximately 0.5 to 1 μm wide. The shear bands contain a high dislocation density. At a given strain these dislocations annihilate and recombine to small angle grain boundaries separating individual grains. Small grains, 8–12 nm diameter, are seen within the shear bands and electron diffraction patterns suggest the misorientation angles between the grains are relatively small. At longer milling times the grain size decreased steadily, consistent with the x-ray results, and the shear bands coalesced. The small angle boundaries were replaced by higher angle boundaries, implying grain rotation, as reflected by the disappearance of texture in the electron diffraction patterns as well as the random orientation of the grains observed from the lattice fringes in HREM. The minimum grain size obtainable by milling has been attributed to a balance between the defect/dislocation structure introduced by the plastic deformation of milling and its recovery by thermal processes (12). It has been found that the minimum grain size induced by milling scales inversely with the melting temperature of the group of fcc metals studied (12). These data are plotted in Figure 1 along with data for bcc and hcp metals (10,11), Si (13,14), and C (graphite (15)). For these data, only the lower melting point (Al, Ag, Cu, Ni) fcc elements show a clear inverse dependence of minimum grain size on melting temperature. The minimum grain size data for the bcc and hcp metals, and for the fcc metals with the higher melting temperatures ($\geq T_M$ for Pd), exhibit essentially constant values with melting temperature. For these elements it appears that $d_{min.}$ is in the order: fcc<bcc<hcp. However, before explanations for the above based on the strain hardening response or other fundamental differences in deformation behavior for the various metallic crystal structures can be considered, it should be pointed out that a number of variables can influence the values of the minimum grain size attained by ball milling. First of all, most of the measurements reported are based on analysis of x-ray diffraction line broadening. Such analysis is subject to difficulty in terms of absolute quantitative values for grain size (16). However, most of the data in Figure 1 are from the Cal Tech group (10,11,12) or from Oleszak and Shingu (17). Both these groups used the approach of Williamson and Hall (18) to estimate the crystallite size and root-mean-square (rms) strain from the x-ray line broadening. In spite of the facts that the mill energies and contamination levels were very different in these studies, remarkable agreement for minimum grain size, d, is found for Al, Cu, Ni, and Fe. The value of d for W is lower (5.5 nm) for Oleszak and Shingu (17) compared to that (9 nm) of Hellstern et al (11). The Cal Tech group (10,11,12) used a high energy shaker mill (Spex 8000) while Oleszak and Shingu (17) used a conventional horizontal low energy ball mill. The metallic impurity level (Fe) from the shaker mill is much larger ($\geq$ 2 at.% Fe) than that from the conventional ball mill (< 0.1 at.% Fe). Conversely, the reported oxygen concentrations were about 4 times higher for powders milled in the conventional mill than for those from the shaker mill. This recent observation

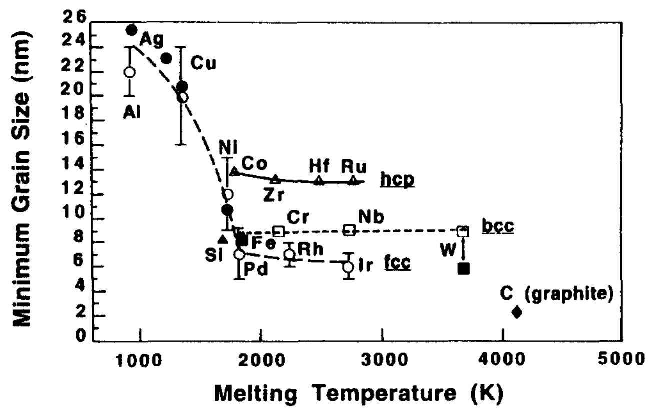

Figure 1. Minimum grain size for nc elements vs. their melting temperatures (open symbols (10,11,12); filled circles (17); Si (13,14); C (15)).

(17) that the minimum nanocrystalline grain sizes for a number of elements milled in a low energy mill are comparable to those milled in a high energy mill (10,11,12) is contrary to conclusions made previously on the minimum grain size in milled TiNi (19). The nanocrystalline grain size was compared after milling TiNi in a high energy Spex shaker mill and a lower energy vibratory mill. After about 10h in the Spex mill the grain size (determined by TEM and x-ray line broadening analysis) decreased to about 4–5 nm. At longer milling times an amorphous structure was observed. Milling for 100h in the vibratory mill resulted in a grain size of about 15 nm. We originally assumed this value represented a saturation to the minimum grain size obtainable in the lower energy mill. However, in light of the work of Oleszak and Shingu (17) it is likely that the $d_{min.}$ was not obtained at 100h in the lower energy mill and that continued milling may have reduced the grain size further. These new results suggest that total strain, rather than milling energy or ball-powder-ball collision frequency, is responsible for determining the nanocrystalline grain size. This is different from amorphization (20,21) or disordering (22) induced by ball milling where it appears the energy and frequency of ball-powder-ball collisions determine the final structures formed in "driven systems". It is, however, consistent with observations of nanocrystallites formed by high strain values using other non-cyclic deformation methods (23–32).

VARIABLES IN THE FORMATION OF NANOCRYSTALLINE MICROSTRUCTURES BY MECHANICAL ATTRITION

Mill Energy

As pointed out above, in elemental powders, similar ultimate nanocrystalline grain sizes have been achieved in high energy shaker mills (10,11,12) and conventional low energy ball mills (17). These results suggest mill energy *per se* is not critical to the final microstructure but of course the kinetics of the process are dependent on the energy, and times for attaining the same microstructure can be several orders of magnitude longer in the low energy mills than in high energy mills.

Milling Temperature

Milling temperature has been observed to affect the rate at which the nanocrystalline structure develops. The milling time at which a given grain size was attained in a TiNi intermetallic compound was a function of milling temperature (19). In this case amorphization occurred at a "critical" grain size of about 4–5 nm so the final nanocrystalline grain size at each milling temperature could not be determined. Shen and Koch (33) also observed smaller nc grain sizes in both Cu and Ni milled at –85°C compared to samples milled at room temperature. For example for Cu $d = 26 \pm 3$ nm for room temperature milling and $d = 17 \pm 2$ nm for milling at –85°C. Evidence for smaller nanocrystalline grain size at low milling temperatures was also obtained on the intermetallic compound CoZr (34).

Alloy Effects

It has been suggested that the ultimate grain size achievable by milling is determined by the minimum grain size that can sustain a dislocation pile-up within a grain and by the rate of recovery during milling (12). To estimate the composition dependence of grain size after milling one may use the formula suggested by Nieh and Wadsworth (35). The minimum distance L between dislocations is $L = 3Gb/\pi(1-\nu)h$ with shear modulus G, Burgers vector b, Poisson ratio ν, and hardness h. If this is indeed related to the formation of a minimum nc grain size, then this nc grain size is inversely proportional to hardness. In fact, a decreasing nc grain size with solute concentration is observed in nc alloy systems which exhibit solid solution hardening such as Cu(Fe) (36,37), Ti(Cu), Nb(Cu), Cu(Ni), and Cu(Co) (37). Also consistent with this is the essentially constant nc grain size in Ni(Co) where hardness does not change significantly (33) and the increased grain size in nc Ni(Cu), Fe(Cu), and Cr(Cu) which exhibit a solid solution softening effect (36).

MECHANISM(S) FOR FORMATION OF NANOCRYSTALLINE MICROSTRUCTURES BY MECHANICAL ATTRITION

While there is some agreement on the phenomenology of the development of a nc grain structure by ball milling as described above, a detailed microscopic model is still lacking, related at least in part to our lack of understanding of the deformation mechanisms in nc materials.

Fecht (9) has summarized the phenomenology of the development of a nanocrystalline microstructure by mechanical attrition into three stages, namely;

stage 1. deformation localization in shear bands containing a high dislocation density.

stage 2. dislocation annihilation/recombination/rearrangement to form a cell/subgrain structure with nanoscale dimensions - further milling extends this structure throughout the sample.

stage 3. the orientation of the grains becomes random, i.e. low angle grain boundaries → high angle grain boundaries — grain boundary sliding, rotation is likely.

From evidence of the atomic level lattice strain and the stored enthalpy as a function of reciprocal grain size (or milling time) it was concluded that two different regimes can be distinguished, i.e. dislocation versus grain boundary deformation mechanisms (9). The lattice strain in the fcc elements studied by Eckert et al (12), (milled in a high energy shaker mill) was found to increase continuously with decreasing grain size and reached a maximum value at the smallest grain size. This is in contrast to the earlier observation on Ru and AlRu which indicated a maximum in strain vs. 1/d (10) and the recent study of Oleszak and Shingu (17) in a low energy mill which also shows a broad maximum in strain vs. 1/d for a number of elements including several fcc elements. The lattice strain values available from the literature are plotted against reciprocal grain size, 1/d, in Figure 2. With the exception of Ru, the data for

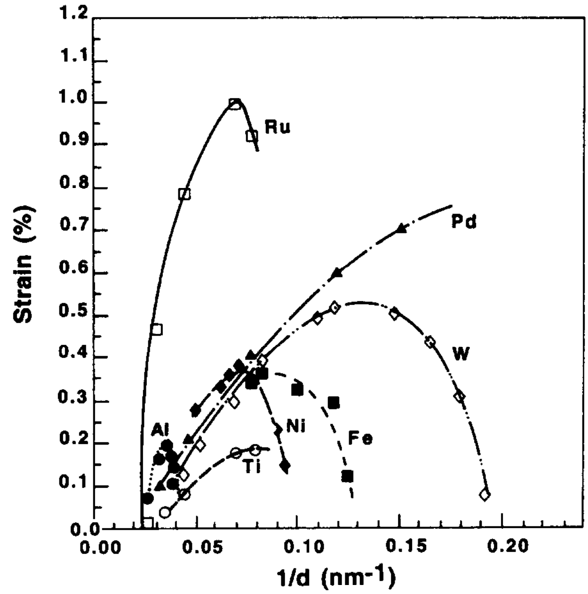

Figure 2. Lattice strain vs. reciprocal grain size. (Al, Ni, Fe, W (17); Ru (10); Pd (12); Ti (56)).

increasing lattice strain with 1/d appear to fall on a common relatively narrow band before decreasing from the maximum strain values. However, these data are from several groups using mills with various energy levels and possible differences in milling temperature. It has been demonstrated (33) that lower milling temperature resulted in larger values for lattice strain for Cu and Ni. However, the data from the low energy mill (17) should be self-consistent and exhibits interesting behavior. That is, the strain rises with decreasing grain size, reaches a maximum, and then decreases to low values for the smallest nanocrystalline grain sizes. A maximum in strain with 1/d was explained previously by either a change in deformation mechanism from dislocation generation and movement to grain boundary sliding, grain rotation (10), or, in brittle intermetallics to fracture after the strain maximum is reached (10,38). The first mechanism is unlikely in this case since the nc grain size continues to decrease after the strain maximum is reached and grain boundary sliding, grain rotation presumably cannot result in grain size reduction. There is no clear evidence at present in the elements for the fracture mechanism which would also presumably result in a finer particulate distribution.

Additional information to help explain the mechanism of nanocrystallite formation comes from measurements of stored enthalpy. Maxima in stored enthalpy vs. 1/d are typically observed (10,12,17). However, the maximum in stored enthalpy is usually found at smaller grain sizes than the strain maximum, as illustrated for W (17) in Figure 3. Here the maximum in strain occurs at d = 8.3 nm while the maximum in stored enthalpy is at d = 5.5 nm. Several suggestions have been offered to explain the maxima in stored enthalpy with 1/d including decreasing strain (10,17) or impurity pick-up during milling (10). The latter is unlikely for the samples from low energy milling (17) where metallic impurity contamination is negligible. It is stated that the stored enthalpy comes mainly from grain boundaries (12,17) and grain boundary strains. Stress relaxation may be responsible for the maxima (17) but, as above, the strain and stored enthalpy maxima don't necessarily coincide.

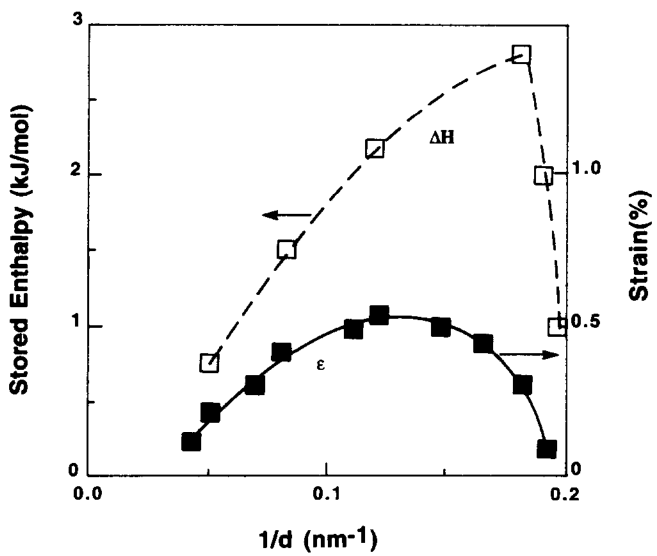

Figure 3. Stored enthalpy (ΔH) and lattice strain (ε) vs. reciprocal grain size for W (17).

The major open question related to the above is: what is the deformation mechanism in materials with nc grain size? In this regard, very interesting *in situ* transmission electron microscopy (TEM) deformation studies have been carried out on nc thin film Au or Ag specimens (39,40,41). Samples with 100 nm grain sizes exhibited the usual deformation mechanisms including extensive dislocation activity, ligament formation, and ductile fracture within a given grain. However, 10 nm grain size samples did not exhibit dislocations within the grains. Instead, evidence was seen for grain boundary rotation and sliding (which can result in 30% strains at the crack tip) as well as opening of small pores at grain boundary triple junctions. Such studies and their analysis should help us better understand the formation of nanocrystalline structures by mechanical attrition.

NANOCRYSTALLINE MICROSTRUCTURES BY MECHANICAL ALLOYING OF ELEMENTAL POWDERS

Most of the studies of the microstructural evolution of a nc structure during ball milling have been carried out on single phase materials such as elements or compounds. However, nanocrystalline grains are also observed during the mechanical alloying (MA) of multicomponent powders. Klassen et al (42) followed the phase formation and microstructural development during MA of Ti and Al powder blends of overall composition $Ti_{25}Al_{25}$. TEM revealed nanocrystalline grains of the partially ordered $L1_2$ phase with a crystallite size of 10–30 nm in the alloy layers at the interface between Ti and Al lamellae at very early stages of the milling process. The alloy phase which develops between the pure powder components consists of nc grains presumably because of the multiple nucleation events and the slow growth which occur at relatively low temperatures (100–200°C above ambient) during milling. Trudeau et al (38) prepared nc FeTi by both MA of elemental Fe and Ti powders and MM of FeTi compound powders. The grain size of MM FeTi steadily decreases with milling time while that for MA Fe/Ti first increases and then decreases to values essentially identical to those for the MM samples.

MECHANICAL CRYSTALLIZATION OF AMORPHOUS ALLOYS

Crystallization has been observed by ball milling certain amorphous materials and nanocrystalline phases often result. In some cases, the crystallization is either clearly due to impurity pick-up changing the composition and therefore crystallization temperature T_x of the alloy (43) or to large local temperature increases to above T_x in energetic mills (eg. 44). However, there are other examples, first cited by Trudeau et al (45), which are not simply explained by the above two effects. Trudeau and co-workers (45,46,47) and Huang et al (48) have observed crystallization induced by ball milling in several Fe-base amorphous alloys in this regard. Explanations offered for this crystallization include: enhancement of diffusion by the milling deformation which allows decomposition of the metastable amorphous phase (49); local temperature rises associated with shear bands due to bending as well as wear processes (48). This phenomenon of crystallization by ball milling in certain amorphous alloys (it is composition dependent) is still not well understood, but may be related to the observation of nanocrystal formation within shear bands of melt spun Al-based amorphous alloys induced by bending the ribbon samples (50). It was suggested this nanocrystallization is due to local atomic rearrangements within the shear bands which exhibit enormous plastic strain. Such atom rearrangement depends on the topological order and the chemical bonding of the amorphous alloy.

PROBLEMS: CONTAMINATION AND POWDER CONSOLIDATION

A serious problem with the milling of fine powders is the potential for significant contamination from the milling media (balls and vial) or atmosphere. If steel balls and containers are used, iron contamination can be a problem. It is most serious for the highly energetic mills, eg. the Spex shaker mill, and depends on the mechanical behavior of the powder being milled as well as its chemical affinity for the milling media. For example, milling Ni to attain the minimum nc grain size in a Spex mill resulted in Fe contamination of 13 at.%, while the Fe contamination in nc Cu similarly milled is only $\leq$ 1 at.% (37). Lower energy mills result is much less, often negligible, Fe contamination. Other milling media, such as tungsten carbide or ceramics, can be used but contamination from such media is also possible. Surfactants (process control agents) may also be used to minimize contamination. Mishurda et al (51) studied the effect of several process control agents on Fe and interstitial element contamination during ball milling of prealloyed $Ti_{50}Al_{48}W_2$ powders. Boric acid and borox were quite effective in reducing Fe contamination (from 1.04 wt.% to 0.44 wt.% and 0.26 wt.%, respectively). Using a "seasoned" vial — i.e. media coated with the alloy powder — resulted in very low values for Fe contamination of 0.06 wt.% (with boric acid) and 0.09 wt.% (with borax). Interstitial contamination can be controlled by milling and subsequent powder handling in a pure inert gas atmosphere with care taken that the milling vial is leak-free during processing. Goodwin and Ward-Close (52) have recently reported on a new production scale mill which can process reactive materials such as Ti-Al alloys with essentially no metallic or interstitial element contamination. Such devices should generate increased interest in the use of mechanical attrition for synthesis of nc materials.

Powder consolidation to theoretical density of nc materials prepared by mechanical attrition without significant coarsening is necessary for many properties eg. mechanical behavior. There is not room in this paper to adequately review the increasing effort in this area. However, a number of successes have been documented by both conventional, eg. hot isostatic pressing (HIP) (53) and dynamic, eg., explosive compaction (54). While theoretical densities have been achieved in many studies, tensile deformation has revealed that interparticle debonding can still be a problem (55). Additional work is needed on this important problem.

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